2×2 Between Groups Factorial ANOVA

When to Use

To examine main effects and interaction relating two independent variables (each with 2 levels) to a single quantitative dependent variable.

Research Hypothesis: Superman movies with different box office performance and budget levels show different patterns of audience ratings. The relationship between box office success ($500M+ vs. Under $500M worldwide) and production budget (Over $230M vs. Under $230M) affects audience ratings, with an interaction effect expected.

Null Hypothesis: There are no main effects of box office category or budget category, and no interaction effect on audience ratings of Superman movies.

Research Design

The design consists of a 2 × 2 table:

	Under $230M Budget	Over $230M Budget
Under $500M BO	n = 4	n = 1
$500M+ BO	n = 2	n = 1

This is a **2 × 2 between-groups factorial design** with two IVs, each having 2 levels.

Variables in the Analysis:

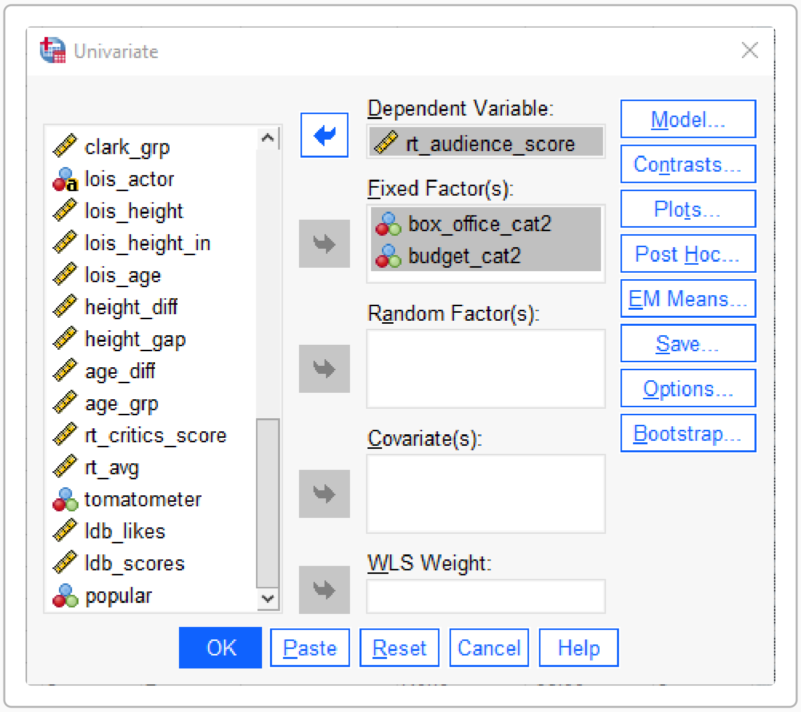
- **IV1 (Box Office Category):** $500M+ (worldwide gross ≥ $500M) vs. Under $500M
- **IV2 (Budget Category):** Over $230M vs. Under $230M
- **DV (Dependent):** Rotten Tomatoes Audience Score

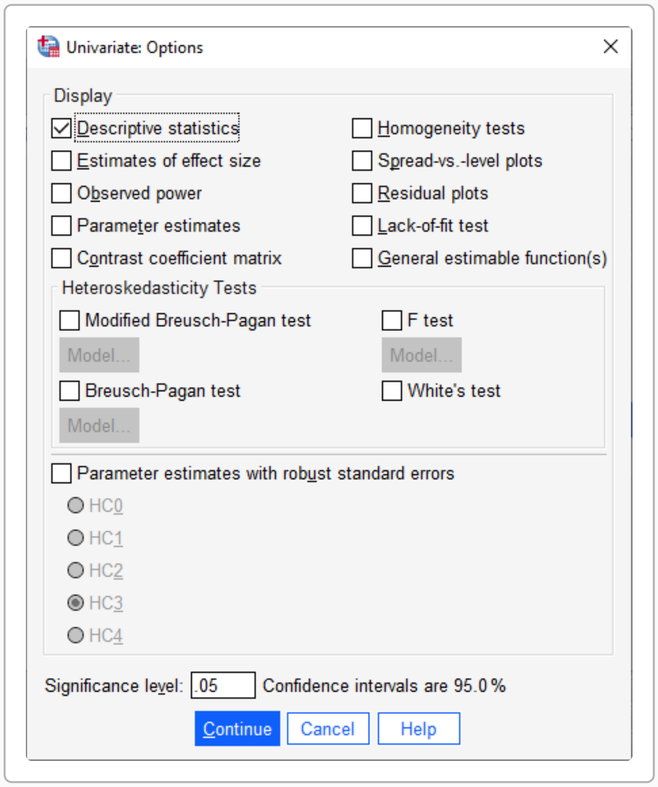
Total Sample: N = 8. The factorial design allows us to examine: the main effect of box office category, the main effect of budget category, and their interaction effect on audience scores.

Procedure

Analyze → General Linear Model → Univariate

- Highlight the "Dependent" variable (be sure it is **quantitative**) and click the arrow to move it to the "Dependent Variable" box
- Highlight both "Factor" variables (IVs, grouping variables) and click the arrow to move them to the "Fixed Factor(s)" box
- "Options" — check that you want "Descriptive Statistics"
- "OK" to run the analysis





In a 2 × 2 factorial design, post hoc tests are usually not necessary because each factor has only 2 levels. Simple effects tests can be used to interpret the interaction effect if needed.

SPSS Syntax

</> SPSS Syntax

```
UNIANOVA rt_audience_score BY box_office_cat2 budget_cat2   ①
  /METHOD=SSTYPE(3)                                         ②
  /INTERCEPT=INCLUDE                                        ③
  /PRINT=DESCRIPTIVE                                        ④
  /CRITERIA=ALPHA(.05).                                     ⑤
```

1. DV "by" both IVs (factors)
2. Use Type III sums of squares (appropriate for unbalanced designs)
3. Include the intercept in the model
4. Get descriptive statistics for each cell
5. Set significance level for the test

SPSS Output

Descriptive Statistics

Dependent Variable: rt_audience_score

box_office_cat2	budget_cat2	Mean	Std. Deviation	N
Under $500M	Under $230M	86.0000	0.0000	4
Under $500M	Over $230M	60.0000		1
	Total	80.8000	11.6276	5
$500M+	Under $230M	82.5000	10.6066	2

Interpreting the Descriptives Table:

- Each cell shows M, SD, and N for that combination
- Row totals show marginal means for each box office group

box_office_cat2	budget_cat2	Mean	Std. Deviation	N
$500M+	Over $230M	75.0000		1
	Total	80.0000	8.6603	3
Total	Under $230M	84.8333	5.0761	6
Total	Over $230M	67.5000	10.6066	2
	Total	80.5000	9.9427	8

- Column totals show marginal means for each budget group
- The grand total is the overall mean

Tests of Between-Subjects Effects

Dependent Variable: rt_audience_score

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	580.636	3	193.545	6.882	.047
Intercept	51842.000	1	51842.000	1843.271	<.001
box_office_cat2	48.091	1	48.091	1.710	.261
budget_cat2	408.091	1	408.091	14.510	.019
box_office_cat2 * budget_cat2	124.455	1	124.455	4.425	.103
Error	112.500	4	28.125		
Total	52535.136	8			
Corrected Total	693.136	7			

a. R Squared = 0.838 (Adjusted R Squared = 0.716)

Interpreting the ANOVA Table

Key rows to focus on:

- **box_office_cat2:** Tests the main effect of box office category on audience ratings
- **budget_cat2:** Tests the main effect of budget category on audience ratings
- **box_office_cat2 × budget_cat2:** Tests the interaction effect — whether the effect of one factor depends on the level of the other
- **Error:** The within-group variance used in all F-test denominator

The **F-statistic** is the ratio of the effect mean square to the error mean square. The **p-value (Sig.)** tells you the probability of obtaining this F if the null hypothesis were true. If $p < .05$, we reject the null hypothesis for that effect.

Effect Size Note: For reporting, the R^2 value shown in the footnote indicates the proportion of variance in audience scores explained by the model (the combination of both factors and their interaction).

Remember: even if SPSS shows $p = .000$, never report $p = .000$ in your results. Instead, report "$p < .001$".

Understanding Main Effects and Interaction

> **! Important**
>
> **Main Effects:** Each factor is tested separately.
>
> - **Box Office main effect:** Compares the average audience score for $500M+ vs. Under $500M movies (averaging across budget categories)
> - **Budget main effect:** Compares the average audience score for Over $230M vs. Under $230M budget movies (averaging across box office categories)
>
> **Interaction Effect:** Tests whether the effect of one factor depends on the level of the other.
>
> - If the interaction is **not significant:** The effects are **additive** — the differences between box office categories are similar regardless of budget category
> - If the interaction **is significant:** The patterns differ — the effect of box office might be stronger for one budget category than the other

Simple Effects and LSD Computations

If the interaction is significant, you can compute the **Least Significant Difference (LSD)** to describe the pattern of the interaction.

Steps for computing LSD:

1. Determine **df(error)** from the ANOVA table
2. Use the t-table to find the critical value of t at p = .05 for this df(error)
3. Get **MSError** from the ANOVA table
4. Apply the LSD formula:

$$d_{LSD} = t_{critical} \times \sqrt{\frac{2 \times MS_{Error}}{n}}$$

5. Compare pairwise mean differences to the LSD value: difference **larger** than LSD → means are significantly different; difference **smaller** than LSD → means are not significantly different

Example: If df(error) = 4 and we want to compare the cell means:

The critical t value at p = .05 for this df would be found in a t-table, then applied to the formula above using the MSError from your output. You can then directly compare mean differences from the descriptive table to your computed d_LSD.

Reporting the Results

Sample APA Write-Up

A 2 x 2 factorial ANOVA was conducted to examine how box office category ($500M+ vs. Under $500M) and budget category (Over $230M vs. Under $230M) affect audience ratings of Superman movies. The cell means were: Under $500M BO/Under $230M Budget (*M* = 86.00, *n* = 4), Under $500M BO/Over $230M Budget (*M* = 60.00, *n* = 1), $500M+ BO/Under $230M Budget (*M* = 82.50, *n* = 2), $500M+ BO/Over $230M Budget (*M* = 75.00, *n* = 1). The main effect of box office category was not significant, $F(1,4) = 1.71$, $p = 0.261$. The main effect of budget category was significant, $F(1,4) = 14.51$, $p = 0.019$. The interaction between box office and budget category was not significant, $F(1,4) = 4.43$, $p = 0.103$, $MSe = 28.12$.

Key Reporting Elements

When reporting 2x2 factorial ANOVA results, be sure to communicate:

- Descriptive statistics for each cell (M, SD, n)
- Both main effects (*F*, *df*, *p*)
- The interaction effect (*F*, *df*, *p*, *MSe*)
- The direction of significant effects
- If the interaction is significant, describe the pattern of cell means
- Whether results support the research hypothesis